

Lecture 11 1. Compare the noise resistance and bandwidth capabilities of Twisted Pair (UTP/STP), Coaxial Cable, and Fiber Optic Cable. Explain how the physical construction of each contributes to its performance in real-world networking environments.

Twisted Pair (UTP / STP):

- **Noise resistance:** Medium (STP higher than UTP)
- **Bandwidth:** Medium to high
- **Construction:** Twisted copper wire pairs (STP has shielding)
- **Use:** LAN, telephone networks

Coaxial Cable:

- **Noise resistance:** High
- **Bandwidth:** High (more than twisted pair)
- **Construction:** Central copper conductor with insulation and metallic shield
- **Use:** Cable TV, broadband internet

Fiber Optic Cable:

- **Noise resistance:** Very high (no EMI)
- **Bandwidth:** Very high
- **Construction:** Glass/plastic fiber carrying light signals
- **Use:** Backbone networks, long-distance communication

Summary:

Fiber optic = best performance, Coaxial = moderate, Twisted pair = basic and low cost.

2. Explain the principle of data transmission in Fiber Optic cables, including the roles of core, cladding, and light refraction. Differentiate between single-mode and multi-mode fiber optics in terms of application scenarios and performance. (4 marks)

Principle of Data Transmission in Fiber Optic Cables

- **How it works:**
Fiber optic cables transmit data using **light signals** instead of electrical signals.
- **Parts of a fiber:**
 1. **Core:** Central part where light travels.
 2. **Cladding:** Surrounds the core and **reflects light back** into the core using **total internal reflection**.
 3. **Light refraction:** Ensures light stays inside the core, even if the cable bends slightly, preventing data loss.

Feature	Single-mode	Multi-mode
Core Size	Very small (~8–10 μm)	Larger (~50–62.5 μm)
Light Path	Straight, single path	Multiple paths (modes)
Bandwidth & Distance	Very high, long-distance	Medium, short-distance
Application	Long-distance telecom, internet backbone	LANs, data centers, shorter links

Fiber optics use **light reflection in the core** for fast, noise-free transmission. Single-mode is for **long distance**, multi-mode for **shorter distance** networks.

3. Analyze the use of Power Line Communication (PLC) as a transmission medium. Discuss its advantages and limitations compared to traditional wired media such as twisted pair and coaxial cable, especially in terms of deployment, data rate, and duplex mode. (4 marks)

Power Line Communication (PLC) as a Transmission Medium

Principle:

PLC uses **existing electrical power lines** to transmit data signals, avoiding the need for separate network cabling.

Advantages of PLC

1. **Easy Deployment:** Uses existing electrical wiring, so no new cables are required.
2. **Cost-Effective:** Reduces installation cost compared to laying twisted pair or coaxial cables.
3. **Convenience:** Can reach areas where other cabling is difficult.

Limitations of PLC

1. **Data Rate:** Lower than twisted pair or coaxial cable, often limited to tens of Mbps.
2. **Noise and Interference:** Power lines are noisy; PLC signals are more prone to interference.
3. **Duplex Mode:** Mostly half-duplex (cannot send and receive at the same time efficiently).
4. **Distance Limitation:** Signal weakens over long distances without repeaters.

Comparison with Twisted Pair / Coaxial

Feature	PLC (Power Line Communication)	Twisted Pair / Coaxial
Deployment	Easy (uses existing wiring)	Requires installing new cables
Data Rate	Low to medium	Medium to high
Duplex Mode	Mostly half-duplex	Full-duplex possible
Noise Resistance	Low (susceptible to electrical noise)	Higher (especially Coaxial with shielding)

4. Discuss why magnetic storage media are still used for large-scale data transfer despite the availability of high-speed internet. Relate your answer to concepts such as bandwidth, cost, feasibility, and security in enterprise environments. (4 marks)

Why Magnetic Storage Media is Still Used for Large-Scale Data Transfer

1. **Bandwidth Limitations:**

Even with high-speed internet, transferring **terabytes of data** can take days or weeks. Magnetic storage (HDDs, tapes) allows **faster bulk transfer** physically.

2. **Cost-Effective:**

Shipping hard drives or tapes is often **cheaper** than paying for extremely high-speed internet connections for large datasets.

3. **Feasibility:**

In some areas, **internet access is limited or unreliable**, making physical storage a practical solution.

4. **Security and Privacy:**

Sensitive enterprise data can be **encrypted and physically transported**, reducing risk of interception compared to online transfer.

✓ **Conclusion:**

Magnetic storage is still used in enterprises because it is **practical, cost-effective, secure, and faster for huge data**, even when high-speed internet exists.

These questions encourage students to think critically about the design, application, and trade offs of different transmission media in modern and legacy networking contexts. write 5 analytical question from this lecture where marks 4 Based on the content of Lecture 11 - Transmission Media, here are five analytical questions (each worth 4 marks) designed to test deeper understanding and real-world application:

1. Explain how the twisted design of UTP and STP cables reduces electromagnetic interference (EMI) and crosstalk. Compare their effectiveness and typical use cases in modern network installations. (4 marks)

Twisted Pair Cables and EMI Reduction

1. How twisting helps:

- Twisting the wires in **UTP (Unshielded Twisted Pair)** and **STP (Shielded Twisted Pair)** causes electromagnetic interference (EMI) from external sources and **crosstalk** between pairs to **cancel out**.
- Each twist changes the orientation of the wires relative to interference, reducing signal corruption.

2. Effectiveness Comparison:

Feature	UTP (Unshielded Twisted Pair)	STP (Shielded Twisted Pair)
EMI & Crosstalk Protection	Moderate	High (has additional shielding)
Cost	Cheaper	More expensive
Installation	Easier, flexible	Slightly harder, needs grounding

3. Typical Use Cases:

- UTP:** Used in **LANs, offices, and homes** where moderate interference exists.
- STP:** Used in **industrial environments or high EMI areas** where extra protection is needed.

2. Describe the structural composition of a coaxial cable and explain why it supports higher bandwidth and better noise immunity compared to twisted pair cables. Provide examples of where coaxial cables are still preferred today. (4 marks)

Structure of Coaxial Cable

1. **Core (Central Conductor):**
 - Usually **copper wire** that carries the electrical signal.
2. **Insulation (Dielectric):**
 - Surrounds the core to **separate it from the outer shield** and maintain signal quality.
3. **Shield (Metallic Braid or Foil):**
 - Protects the signal from **electromagnetic interference (EMI)** and reduces **crosstalk**.
4. **Outer Jacket:**
 - Plastic layer that **protects the cable physically**.

Why Coaxial Supports Higher Bandwidth and Noise Immunity

- **Shielding + single conductor design** → reduces external interference more than twisted pair.
- **Consistent spacing** between core and shield → supports **higher frequency signals** → higher bandwidth.
- **Less crosstalk** compared to twisted pair.

Examples of Current Use

- Cable TV (CATV)
- Broadband internet in homes (coaxial broadband)
- CCTV camera connections

3. Analyze Power Line Communication (PLC) as an alternative data transmission medium. Discuss its operational principles, advantages in terms of infrastructure reuse, and limitations related to data rate and duplex communication. (4 marks)

3. Power Line Communication (PLC)

Operational Principles:

- PLC uses **existing electrical power lines** to transmit data signals along with electricity.
- Data is modulated onto the AC line using special PLC adapters.

Advantages:

- **Infrastructure Reuse:** No need to lay new cables; uses existing power wiring.
- **Easy Deployment:** Works in homes, offices, and industrial environments.

Limitations:

- **Data Rate:** Lower than twisted pair or coaxial (tens of Mbps).
- **Duplex Communication:** Mostly half-duplex (cannot send and receive efficiently at the same time).
- **Noise:** Power lines are noisy, can interfere with data signals.

2. Compare single-mode and multi-mode fiber optic cables in terms of light transmission, distance capabilities, and typical applications. Explain why single-mode fiber is preferred for long-distance communication. (4 marks) .

Feature	Single-mode	Multi-mode
Light Transmission	Single straight path	Multiple paths (modes)
Distance Capability	Very long (tens of km)	Short (up to 2 km)
Bandwidth	Very high	Medium
Applications	Telecom backbone, long-distance internet	LANs, data centers, short-distance connections

Why Single-mode is preferred for long distance:

- Light travels straight, so **less signal dispersion** and **higher bandwidth** over long distances.

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3. Evaluate the scenarios where physical magnetic media (e.g., tapes, disks) are still used for data transfer despite the availability of high-speed networks. Consider factors such as data volume, security, cost, and network limitations. (4 marks).

Use of Magnetic Media for Data Transfer

Scenarios where still used:

- **Data Volume:** Transferring very large datasets (terabytes) is faster physically than via internet.
- **Security:** Sensitive enterprise data can be **encrypted and physically transported**, reducing hacking risk.
- **Cost:** Cheaper than upgrading to ultra-high-speed internet for massive transfers.
- **Network Limitations:** Areas with slow or unreliable internet rely on physical media.

Examples: Hard drives, magnetic tapes, portable disks.

Lecture 14 1. Compare Circuit Switching and Packet Switching in terms of resource allocation, delay characteristics, and suitability for different types of applications (e.g., voice vs. data). Provide an example where each method is optimally used. (4 marks)

Circuit Switching vs Packet Switching

Feature	Circuit Switching	Packet Switching
Resource Allocation	Dedicated path for entire session	Shared resources; packets sent independently
Delay	Constant once established, but initial setup delay	Variable delay due to queuing and congestion
Application Suitability	Voice calls (continuous, real-time)	Data transfer, emails, web browsing (burst traffic)
Example	Traditional telephone network	Internet, LANs

✓ **Summary:** Circuit switching is ideal for **real-time voice**, packet switching is efficient for **data communication**.

2. Explain why Message Switching is no longer widely used in modern networks. Discuss its limitations in terms of storage requirements, latency, and suitability for real-time communication compared to Packet Switching. (4 marks)
3.
 - **Message Switching** stores the entire message at a switching node before forwarding it to the next node.
 - This requires **large storage** at each node to hold complete messages, which is expensive and inefficient.
 - Because the whole message must be received before forwarding, it causes **high latency** (slow delivery).
 - It is **unsuitable for real-time communication** like video calls or live streaming, where fast delivery is required.
7.
 - **Packet Switching** splits messages into smaller packets, reducing storage needs, lowering delay, and allowing real-time data transfer.
 - Therefore, **Message Switching is rarely used** in modern networks due to its storage demands, delay, and poor real-time performance.
4. Describe the three phases of Circuit Switching and explain how this approach ensures reliable communication for voice applications. Contrast this with the connectionless nature of Packet Switching in the Internet.

3. Circuit Switching Phases and Reliability

Three Phases of Circuit Switching:

1. **Establishment Phase:** A dedicated path (circuit) is set up between the sender and receiver before data transfer begins.
2. **Data Transfer Phase:** Data flows continuously along the dedicated path with minimal delay.
3. **Termination Phase:** The circuit is released after communication ends, freeing resources.

Reliability for Voice Applications:

- Provides a **constant, reserved bandwidth**, ensuring **no delay or jitter**, which is ideal for **voice calls**.

Contrast with Packet Switching:

- Packet switching is **connectionless**: data is sent in packets without a dedicated path.
- Packets may take different routes, causing **variable delay and out-of-order arrival**, which is less suitable for real-time voice without extra mechanisms.

4. Analyze how Packet Switching improves line efficiency through multiplexing and prioritization. Discuss how Quality of Service (QoS) can be implemented in a packet-switched network using priority-based forwarding. (4 marks)

Packet Switching, Line Efficiency, and QoS

Line Efficiency Improvement:

1. Packet switching allows multiple users to share the same line via **multiplexing**, using bandwidth efficiently.
2. Packets from different sources can be interleaved, reducing idle time and improving **line utilization**.

Quality of Service (QoS):

1. QoS ensures certain packets (like voice or video) get priority over less urgent packets.
2. Implemented using **priority-based forwarding**, where routers handle high-priority packets first.
3. This reduces **delay and jitter** for critical applications while still supporting other traffic.
4. Compare connectionless and connection-oriented switching approaches in the context of Packet Switching. Explain the role of forwarding tables and pre-established circuits, and discuss the trade-offs in terms of setup delay and reliability.

Connectionless vs Connection-Oriented Packet Switching

Connectionless (Datagram) Switching:

1. Each packet is sent independently with its **destination address**.
2. **Forwarding tables** at routers determine the next hop for each packet.
3. No setup is needed, so **low setup delay**, but packets may arrive out-of-order.

Connection-Oriented (Virtual Circuit) Switching:

1. A **virtual circuit** is established before sending packets.
2. All packets follow the **same path**, ensuring **in-order delivery**.
3. **Pre-established circuits** require setup time but improve **reliability**.

Trade-offs:

- Connectionless: fast to start, less reliable, out-of-order delivery possible.
- Connection-oriented: setup delay, more reliable, ordered delivery, suitable for predictable traffic.

Lecture 16 1. Compare the Stop-and-Wait and Sliding Window flow control mechanisms in terms of efficiency, resource utilization, and suitability for different network conditions. Provide a scenario where each method would be preferred. (4 marks)

1. Stop-and-Wait vs Sliding Window Flow Control

Stop-and-Wait:

1. Sends **one frame at a time** and waits for acknowledgment (ACK) before sending the next.
2. **Efficiency:** Low for long-distance or high-latency networks because the sender is idle while waiting.
3. **Resource Utilization:** Minimal; needs little buffer.
4. **Scenario:** Simple LAN or short-distance links with low traffic.

Sliding Window:

1. Allows **multiple frames to be sent** before waiting for ACKs.
2. **Efficiency:** High, as the sender can transmit continuously.
3. **Resource Utilization:** Requires larger buffers at sender and receiver.
4. **Scenario:** High-speed networks like WANs or satellite links where round-trip delay is significant.

3. Explain how Automatic Repeat Request (ARQ) mechanisms handle error control in the data-link layer. Differentiate between Stop-and-Wait ARQ, Go-Back-N ARQ, and Selective Repeat ARQ in terms of retransmission strategy and buffer requirements. (4 marks)

2. ARQ Mechanisms for Error Control

Automatic Repeat Request (ARQ):

- Detects errors using **ACK/NACK** and retransmits lost or corrupted frames.

Stop-and-Wait ARQ:

- Retransmits **only one frame at a time** if ACK not received.
- Low buffer requirement, simple but **inefficient for high-latency links**.

Go-Back-N ARQ:

- Sender can transmit **N frames** without waiting.
- If an error occurs, **all frames from the error onward are retransmitted**.

- Requires **larger sender buffer**, simple receiver buffer.

Selective Repeat ARQ:

- Sender retransmits **only erroneous frames**.
- Receiver must buffer out-of-order frames.
- **More efficient**, but **higher memory usage and complexity**.

3. Describe the role of positive ACK, negative ACK (NACK), and timeout in error control protocols. Explain how these mechanisms work together to ensure reliable data transmission over an unreliable medium. (4 marks)

Role of ACK, NACK, and Timeout

1. **Positive ACK (ACK):** Receiver confirms **successful receipt** of a frame.
 2. **Negative ACK (NACK):** Receiver indicates a **frame was corrupted** or missing.
 3. **Timeout:** If sender doesn't receive ACK within a certain time, it **retransmits** the frame.
 4. **Together:** These mechanisms ensure **reliable data transfer**, even over an unreliable channel, by detecting errors and guaranteeing all frames are delivered correctly.
5. **Analyze the trade-offs between Go-Back-N ARQ and Selective Repeat ARQ in terms of bandwidth efficiency, complexity, and memory usage at the receiver. In which network conditions would Selective Repeat ARQ be more advantageous? (4 marks)**

4. Go-Back-N vs Selective Repeat ARQ

Go-Back-N ARQ:

- **Bandwidth efficiency:** Lower (retransmits multiple frames even if only one error occurs).
- **Complexity:** Simple to implement.
- **Memory:** Receiver needs minimal buffer.

Selective Repeat ARQ:

- **Bandwidth efficiency:** Higher (only erroneous frames retransmitted).
- **Complexity:** Higher due to out-of-order frame management.
- **Memory:** Receiver needs buffer for out-of-order frames.
- **Best Network Conditions:** High-latency or error-prone links where retransmitting many frames is costly

5. Illustrate with a diagram or example how the Sliding Window protocol can be combined with ARQ to achieve both flow control and error control simultaneously. Explain how window size affects performance.

Sliding Window + ARQ (Flow Control + Error Control)

How it Works:

1. The **Sliding Window protocol** lets the sender transmit multiple frames before needing an acknowledgment, which controls the **flow of data**.
2. **ARQ (Automatic Repeat Request)** detects errors or lost frames and retransmits them, ensuring **reliable delivery**.
3. Together, they allow **continuous, reliable transmission** without overwhelming the receiver.

Example Scenario:

- **Sender Window Size = 4**
- **Frames to send: F1, F2, F3, F4, F5, F6...**
- **Receiver Window Size = 4**

Sender Window: [F1 F2 F3 F4] --> sent

Receiver expects: [F1 F2 F3 F4]

If F2 is lost or corrupted:

Receiver sends NACK(F2)

Sender retransmits F2

Then Sender slides window forward:

New Sender Window: [F3 F4 F5 F6] --> continue sending

Lecture 17 1. Compare and contrast Single-bit, Multiple-bit, and Burst errors in terms of their causes, detection difficulty, and impact on data integrity. Provide an example scenario where each type might occur in a communication system. (4 marks)

Error Type	Cause	Detection Difficulty	Impact	Example Scenario
Single-bit Error	Noise or interference flips one bit	Easy to detect using parity	Minor, affects only 1 bit	A single bit flipped during satellite transmission
Multiple-bit Error	Several bits flipped in a message	Harder to detect with simple parity	Can corrupt meaningful data	Electrical noise causing 2–3 bits to flip in a LAN packet
Burst Error	Many consecutive bits corrupted	Difficult to detect with basic methods	Major, can destroy entire frame	Lightning strike on a long cable or fading in wireless link

Conclusion:

- Single-bit errors are easiest to detect and correct.
- Burst errors are most dangerous and require advanced error control techniques.

2. Explain how Parity Check works for error detection, including even and odd parity schemes. Discuss its limitations and explain why it is ineffective for detecting multiple-bit or burst errors. (4 marks)

How It Works:

1. Adds **one parity bit** to a group of bits for error detection.
2. **Even Parity:** Number of 1s in the message + parity bit = even.
3. **Odd Parity:** Number of 1s in the message + parity bit = odd.

Limitations:

- Can only **detect an odd number of bit errors**.
- Fails to detect **even number of errors** or burst errors.

Example:

- Message: 1010
- **Even parity:** parity bit = 0 → 10100 (even number of 1s)
- If 2 bits flip → parity still even → **error not detected**

Conclusion: Parity check is **simple but not reliable** for multiple-bit or burst errors.

3. Describe the Cyclic Redundancy Check (CRC) method for error detection. Explain the role of the polynomial divisor, codeword formation, and the division process at both sender and receiver. Why is CRC more robust than parity check? (4 marks)

1. The sender chooses a **generator polynomial** (divisor) agreed upon with the receiver.
2. Original data (message) is appended with **redundant bits (CRC code)** so that the total message (codeword) is divisible by the polynomial.
3. **Division Process at Sender:**
 - Treat the message as a binary number.
 - Divide by the generator polynomial using **modulo-2 division**.
 - Remainder becomes the **CRC bits** and is appended to the message.
4. **Division Process at Receiver:**
 - Receiver divides the received codeword by the same polynomial.
 - If remainder = 0 → no error; otherwise → error detected.

Why CRC is more robust than parity:

- Can detect **single-bit, multiple-bit, and burst errors**.
- Parity can only detect **single-bit or odd-number errors**.

Example:

- Message: 1011
- Generator polynomial: 1101
- CRC computed and appended → sent codeword
- Receiver performs same division → remainder = 0 → correct

4. Differentiate between Backward Error Correction (BEC) and Forward Error Correction (FEC). Discuss the scenarios where each method is preferred, considering factors such as transmission medium, cost, and latency. (4 marks)

Type	How it Works	Latency	Cost / Complexity	Preferred Scenario
Backward Error Correction (BEC)	Receiver detects error → requests retransmission (ARQ)	Higher, due to retransmission	Low complexity	Reliable, low-error links like LAN

Type	How it Works	Latency	Cost / Complexity	Preferred Scenario
Forward Error Correction (FEC)	Sender adds redundant bits → receiver corrects errors without asking	Low latency	Higher complexity and cost	Noisy or high-delay links like satellite or deep-space communication

Conclusion:

- BEC is **cheaper and simpler** but slower.
- FEC is **faster and robust**, suitable for high-latency or unreliable channels.

5. Using the formula $2^r \geq m+r+1$, calculate the minimum number of redundant bits r required for error correction of a 7-bit ASCII character. Explain the logic behind this inequality in the context of Hamming distance and error correction capability. (4 marks)

Hamming Code – Minimum Redundant Bits

Given Formula:

$$2^r \geq m+r+1$$

- $m = 7$ (number of data bits for ASCII)
- Find r (redundant bits)

Step-by-Step Calculation:

1. Try $r = 3 \rightarrow 2^3 = 8$; $m+r+1 = 7+3+1 = 11$
 $8 < 11$ ✗
2. Try $r = 4 \rightarrow 2^4 = 16$; $m+r+1 = 7+4+1 = 12$
 $16 \geq 12$ ✓

✓ Minimum $r = 4$ redundant bits

Logic of Inequality:

- 2^r = number of distinct patterns the redundant bits can represent
- $m+r+1$ = total bits + 1 (to detect/correct single-bit error)
- Ensures **Hamming distance = 3**, allowing **single-bit error correction**

Lecture 24 1. Explain the role of TCP flags (SYN, ACK, FIN, RST, PSH, URG) in connection management, data transfer, and error handling. Illustrate how these flags are used in a typical TCP three way handshake and connection termination. (4 marks)

Common TCP Flags:

- **SYN:** Initiates a connection (synchronize sequence numbers).
- **ACK:** Acknowledges received data or SYN.
- **FIN:** Terminates a connection.
- **RST:** Resets a connection in case of errors.
- **PSH:** Pushes data to the receiving application immediately.
- **URG:** Marks urgent data that needs immediate attention.

TCP Three-Way Handshake (Connection Setup):

1. Client → Server: **SYN** (request to connect, initial sequence number).
 2. Server → Client: **SYN + ACK** (acknowledge client SYN, send server sequence number).
 3. Client → Server: **ACK** (acknowledge server SYN).
- ✓ Connection established.

Connection Termination:

1. One side sends **FIN** → request to close connection.
 2. Other side sends **ACK** → acknowledge FIN.
 3. Other side sends **FIN**, then first side sends **ACK**.
- ✓ Connection closed gracefully.

.2. Describe TCP's congestion control mechanism, including Slow Start, Additive Increase/Multiplicative Decrease, and Timeout React. Explain how the window size is adjusted in response to network conditions and packet loss. (4 marks)

TCP Congestion Control

Mechanisms:

1. **Slow Start:** Begin with small congestion window (cwnd) and increase exponentially until threshold.
2. **Additive Increase/Multiplicative Decrease (AIMD):**
 - Gradually increase window size (additive) when no loss.
 - Halve window size (multiplicative) when packet loss occurs.
3. **Timeout React:** If timeout occurs, reset cwnd to 1 and start slow start again.

Window Size Adjustment:

- Window grows with successful ACKs.
- Shrinks rapidly on packet loss to avoid congestion.

.3. Analyze how TCP ensures reliable data delivery through sequence numbers, acknowledgements, and retransmission timers. Discuss what happens when segments arrive out of order or are lost, and how duplicate segments are handled. (4 marks)

3. TCP Reliable Data Delivery

1. Each segment has a **sequence number** for ordering.
2. Receiver sends **ACK** for successfully received segments.
3. **Retransmission Timer**: If ACK not received before timeout, segment is retransmitted.
4. **Out-of-Order Segments**: Buffered until missing segments arrive.
5. **Lost Segments**: Detected by missing ACKs → retransmitted.
6. **Duplicate Segments**: Ignored if sequence number already acknowledged.

✓ Ensures **reliable, ordered, and loss-free delivery**.

.4. Compare the functions of the four TCP timers: Keep-alive, Retransmission, Persist, and Timed-Wait. Explain how each timer contributes to connection stability, flow control, and crash recovery. (4 marks)

TCP Timers

Timer	Function
Keep-alive	Detect idle or dead connections, sends probe to check if peer is alive.
Retransmission	Retransmits segment if ACK not received in time (error recovery).
Persist	Prevents deadlock when receiver window = 0; probes for window updates.
Timed-Wait	Waits before fully closing connection to handle delayed segments.

✓ All timers contribute to **connection stability, flow control, and crash recovery**.

.5. Discuss how TCP multiplexing works and its impact on server performance. Explain the benefits and potential drawbacks of keeping TCP sessions open for longer periods, especially in the context of HTTP/1.1 and modern web applications. (4 marks)

TCP Multiplexing and Session Management

1. **TCP Multiplexing:** Allows multiple TCP connections on a single server using **ports**.
2. **Impact on Performance:**
 - Efficient resource usage → one server handles many clients.
 - Overhead increases if too many sessions are open.
3. **Keeping Sessions Open (HTTP/1.1 & modern web):**
 - **Benefits:** Reduces latency, avoids repeated handshakes, improves throughput.
 - **Drawbacks:** Consumes memory and file descriptors, can affect scalability under heavy load.

✓ Multiplexing improves performance but requires careful session management.

Lecture 24 1. Compare and contrast UDP and TCP in terms of reliability, connection management, header complexity, and suitability for different types of applications. Provide examples where UDP is preferred over TCP. (4 marks)

Feature	TCP	UDP
Reliability	Reliable: ensures delivery, order, retransmission	Unreliable: no guarantee of delivery/order
Connection Management	Connection-oriented: 3-way handshake	Connectionless: no handshake
Header Complexity	Large header (20 bytes)	Small header (8 bytes)
Suitability / Example	File transfer, email, web (HTTP, FTP)	VoIP, video streaming, online games

When UDP is preferred:

- Real-time or fast applications where **low latency** is more important than guaranteed delivery.

.2. Explain why UDP is considered an “unreliable” protocol and discuss the implications of this unreliability in real-time applications such as VoIP and video streaming. How does the underlying IP protocol’s “best-effort” delivery mitigate some of these issues? (4 marks)

Why UDP is “Unreliable”

1. UDP does not perform **acknowledgment, retransmission, or sequencing**.
2. If packets are lost, duplicated, or arrive out of order → application handles it.
3. **Implications for real-time apps:**
 - VoIP, video, gaming tolerate some loss but cannot wait for retransmission → low latency critical.
4. **IP best-effort delivery:**
 - IP attempts to deliver packets but offers no guarantee.
 - UDP relies on this for basic delivery, minimizing overhead.

✓ UDP’s simplicity improves **speed and low-latency performance** at the cost of reliability.

.3. Analyze the UDP header structure and explain the purpose of each field (Source Port, Destination Port, Length, Checksum). Why is the checksum optional in IPv4, and what are the security or reliability risks of disabling it? (4 marks)

UDP Header Structure

Field	Purpose
Source Port	Identifies sending application (optional if not needed)
Destination Port	Identifies receiving application
Length	Total length of UDP header + data
Checksum	Detects errors in header and data (optional in IPv4, mandatory in IPv6)

.Why checksum is optional in IPv4:

- Some systems skip it for **speed optimization** on trusted networks.

Risks of disabling checksum:

- Corrupted data may go undetected → **reliability and security issues**.

.4. Discuss the role of UDP in query-based communications and protocols like DNS and SNMP. Why is UDP's stateless and connectionless nature advantageous in these contexts? (4 marks)

UDP in Query-Based Protocols

- Protocols like **DNS and SNMP** use UDP.
- **Stateless and connectionless nature** is advantageous because:
 1. No need to establish a connection → faster queries.
 2. Server can handle many simultaneous requests without memory overhead.
 3. Lightweight communication suitable for small, simple queries.

✓ UDP is ideal for quick, repeated query-response interactions.

.5. Evaluate the trade-offs between using UDP and TCP for multimedia streaming. Consider factors such as latency, bandwidth efficiency, packet loss tolerance, and congestion control.

Factor	TCP	UDP
Latency	Higher due to connection & retransmission	Low latency, better for real-time media
Bandwidth Efficiency	Less efficient (overhead + retransmissions)	High efficiency, minimal overhead
Packet Loss Tolerance	Low tolerance: retransmits lost packets	High tolerance: can skip lost frames
Congestion Control	Built-in: reduces speed when network congested	None: can maintain speed but risk overload

Conclusion:

- **UDP** is preferred for live streaming, VoIP, online gaming.
- **TCP** is preferred for downloads, file transfer, or when reliability is critical.

Lecture 28 1. Compare and contrast DNS and HTTP in terms of their transport layer protocols (UDP vs. TCP), communication models, and roles in the Internet infrastructure. Explain why DNS typically uses UDP despite being a critical service. (4 marks)

Feature	DNS	HTTP
Transport Protocol	UDP (mostly) for queries, TCP for zone transfers	TCP (reliable connection)
Communication Model	Query-response, stateless	Request-response, connection-oriented
Role in Internet	Resolves domain names to IP addresses	Transfers web content (web pages, files)
Why DNS uses UDP	Faster, lower overhead; queries are small, stateless; reliability can be handled by retries	N/A

Note: DNS is fast and lightweight, critical for resolving addresses quickly.

.2. Explain the process of email transmission using SMTP, POP3, and IMAP. Discuss the roles of User Agents, Message Transfer Agents, and mail servers, and analyze the differences between POP3's delete and keep modes. (4 marks)

Email Transmission (SMTP, POP3, IMAP)

Components:

- **User Agent (UA):** Email client (Outlook, Gmail) used by user.
- **Message Transfer Agent (MTA):** Server that transfers mail (SMTP).
- **Mail Servers:** Store incoming messages until user retrieves them.

Process:

1. **SMTP:** Used to **send emails** from client to server or between servers.
2. **POP3:** Retrieves emails from server; can **delete emails** after download or keep them on server.
3. **IMAP:** Retrieves emails while **keeping them on server**, allows multiple device access.

POP3 Delete vs Keep Modes:

- **Delete Mode:** Messages removed from server after download → offline access only.
- **Keep Mode:** Messages remain on server → access from multiple devices possible.

.3. Describe how FTP uses out-of-band control and data connections (ports 20 and 21). Discuss the advantages and disadvantages of this approach compared to using a single connection for both control and data, as in HTTP. (4 marks)

FTP Control and Data Connections

- **Control Connection (Port 21):** Used for commands (login, navigation).
- **Data Connection (Port 20):** Used for file transfers (upload/download).

Advantages:

- Separation of control and data allows **simultaneous command processing and file transfer**.
- Easier to manage large files and complex commands.

Disadvantages:

- Two connections are more complex to firewall/NAT.
- Less straightforward than single connection (HTTP).

.4. Analyze the evolution of HTTP from version 1.0 to 1.1 in terms of connection persistence. Explain the impact of persistent and non-persistent connections on web performance, server load, and latency. (4 marks)

Feature	HTTP 1.0	HTTP 1.1
Connection	Non-persistent (new TCP for each request)	Persistent by default (keep connection open)
Impact on Performance	Higher latency due to multiple TCP handshakes	Lower latency, fewer handshakes
Server Load	More TCP connections → higher load	Fewer connections → better efficiency
Bandwidth Usage	Higher (repeated handshake overhead)	Lower (reused connection)

✓ Persistent connections improve speed and reduce server overhead.

.5. Explain why HTTP is considered a stateless protocol and discuss the mechanisms (such as cookies and sessions) used to maintain state in web applications. How does statelessness affect scalability and security?

HTTP is Stateless

- **Stateless:** Each HTTP request is **independent**, server does not remember previous requests.
- **Maintaining State:**
 - **Cookies:** Small data stored on client, sent with requests.
 - **Sessions:** Server-side storage linked to client via session ID.
- **Impact:**
 - **Scalability:** Easier to scale servers since no state is stored by default.
 - **Security:** Statelessness reduces risk of session hijacking, but cookies/sessions introduce security considerations (e.g., CSRF, XSS).

✓ Statelessness simplifies server design but requires extra mechanisms for persistent user experience.